

Airbnb Mobile App Heuristic Evaluation

An expert usability inspection of the Airbnb iOS mobile app, focused on the guest booking experience.

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Academic UX Case Study

Note: This case study was completed as part of CCPS 613: Human-Computer Interaction and has been reformatted from the original lab submission into a portfolio-style UX report.

Executive Summary

This usability inspection assessed the Airbnb iOS mobile app using Nielsen's 10 heuristics, focusing specifically on the guest booking experience. Fourteen unique usability issues were identified across the search flow, listing comparison, booking steps, reservation management, and messaging. Most issues were minor, but four were major and directly affected user control, error prevention, or clarity in critical tasks. Prioritization based on severity and estimated fix effort highlighted P13 (accidental cancellation risk), P3 (loss of scroll position), P14 (unclear message errors), and P11 (unexpected back-navigation) as the highest-value improvements to address. Implementing the recommended changes would significantly strengthen efficiency, user confidence, and error recovery within the Airbnb booking experience.

1. System Overview

The evaluated system is the Airbnb mobile app (iOS). The scope of this inspection focuses on the guest-side booking journey, including:

- Searching for accommodations
- Applying filters
- Browsing and comparing listings (map + list)
- Advancing through the booking flow up to the payment screen
- Modifying or canceling reservations
- Contacting hosts through the messaging interface

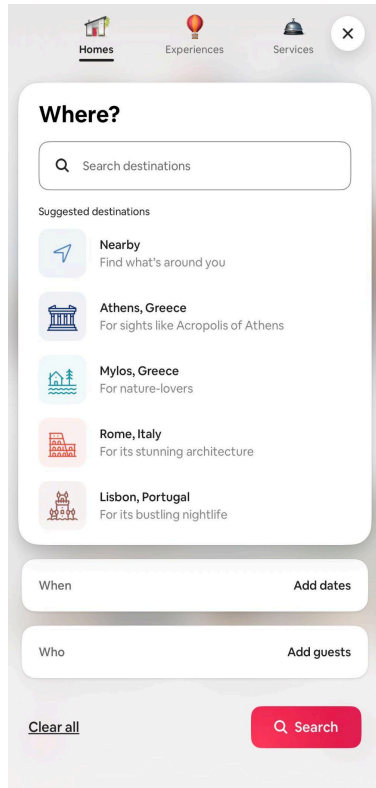


Figure 1: Guest booking search interface showing the evaluation scope.

This guest-side flow was selected because it represents the most common real-world usage pattern and provides enough interaction depth to surface a diverse range of usability issues.

2. Methodology

A heuristic evaluation was conducted following Nielsen's 10 usability heuristics. The process included a familiarization phase, a task-driven walkthrough, independent identification of usability issues, and the application of severity ratings (0–4).

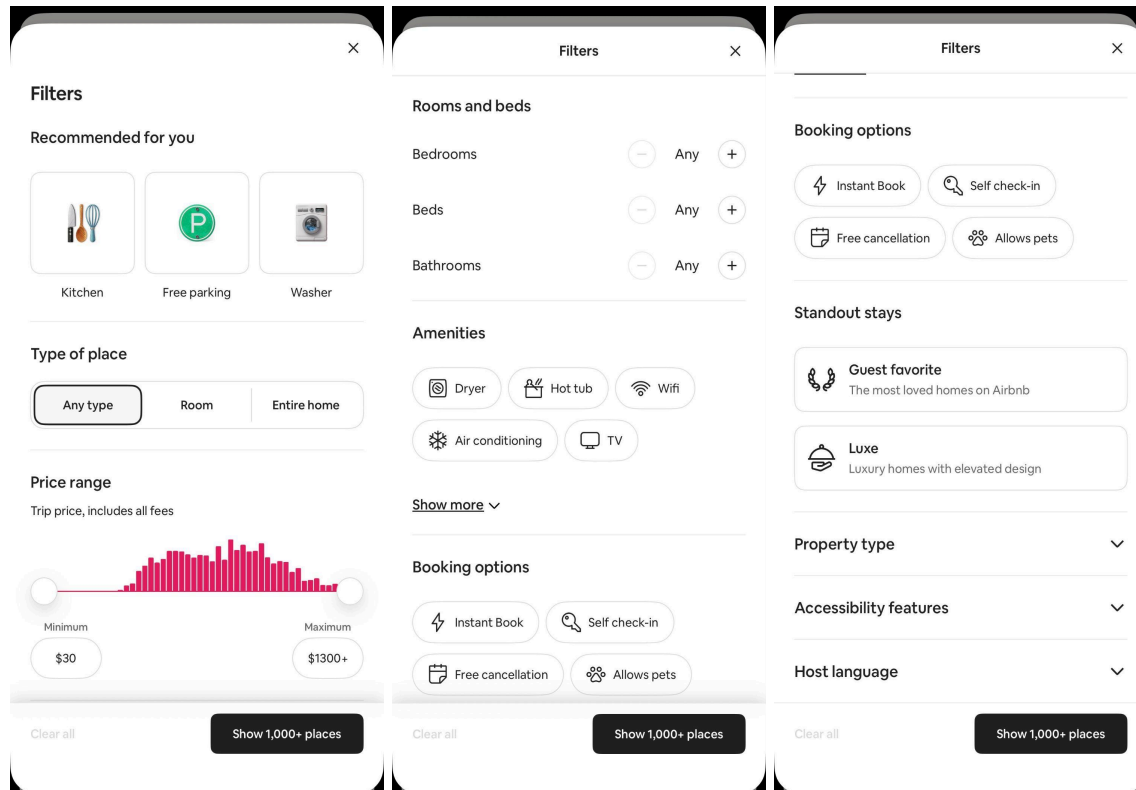


Figure 2: Filter interface explored during the familiarization phase.

This familiarization step helped establish the app's navigation patterns, filter behavior, and mobile interaction flow before identifying heuristic violations.

3. User Persona

A tech-comfortable traveler, age 25–45, who plans and books short-term rentals primarily through a mobile device. This persona relies on quick scanning, visual clarity, and trustworthy information while planning trips.

4. Task List

The evaluation followed these five core tasks:

1. Search for accommodations using filters
2. Compare multiple listings and read reviews
3. Proceed through the booking flow up to payment

4. Modify or cancel an existing reservation
5. Contact a host using in-app messaging

These tasks reflect the typical user journey in the Airbnb mobile app.

5. Heuristic Violations Overview

5.1) Summary Statistics

- Total issues identified: 14
- Severity Breakdown:
 - Cosmetic: 2
 - Minor: 8
 - Major: 4
 - Catastrophic: 0
- Most frequently violated heuristics:
 - Visibility of System Status
 - User Control & Freedom
 - Error Recovery

5.2) Heuristic Violations Table

Heuristic Evaluation Summary Table

Heuristic ↓ / Task →	Task 1: Search + Filters	Task 2: Compare Listings	Task 3: Booking Flow	Task 4: Modify / Cancel Reservatio n	Task 5: Contact Host
1. Visibility of System Status	Filter updates show no loading indicator; results refresh silently	Photo carousel lacks “photo count” indicator	Price breakdown loads slowly with no feedback	Trips page sometimes loads with no progress indicator	Message status (“sent/delivered”) not clearly visible
2. Match Between System and Real World	Some filter icons (like “Type of	Review category labels appear	Fee terminology (“Service	Cancellation policy uses vague labels	Message timestamps are tiny and

	place”) are ambiguous	inconsistently	fee”, “Before taxes”) may be unclear	(“Moderate”)	hard to read
3. User Control & Freedom	Hard to undo accidental filter selections	Switching between listings loses scroll position	Back button sometimes exits booking instead of previous step	Cancelling changes requires scrolling to bottom of modal	No quick way to delete drafted messages
4. Consistency & Standards	Filter panels use inconsistent label ordering	Amenity layouts vary between listings	Date picker differs between search flow and booking flow	“Change reservation” options appear in different places	Messaging UI uses nonstandard chat conventions
5. Error Prevention	Easy to mis-tap dates due to small touch targets	Wishlist/Share buttons are small and easy to mis-tap	Guest count step can be skipped accidentally	Cancellation options sit close together → tap error risk	Easy to hit “Send” accidentally due to button placement
6. Recognition Rather Than Recall	Filter labels like “Essentials” require prior knowledge	Listing info scattered across long scroll → hard to recall	Booking hides key info behind collapsible sections	Past trips hidden under small “Trips” tab	“Contact Host” buried deep in listing page
7. Flexibility & Efficiency of Use	No quick reset for filters beyond “Clear all”	Hard to compare listings side-by-side	No autofill for common guest info	No shortcut to frequently modified trips	No saved message templates for common questions
8. Aesthetic & Minimalist Design	Filter panel dense with text and toggles	Listing pages visually cluttered and long	Booking page contains many fee blocks	Trips details are text-heavy	Messaging thread repeats host info unnecessarily

					y
9. Help Users Recognize, Diagnose, Recover From Errors	No clear feedback if location search is invalid	Listing load errors show vague messages	Card decline error message is generic	Errors modifying reservations lack actionable guidance	Message send failures lack explanation
10. Help & Documentation	No in-flow help for using advanced filters	Amenities sometimes lack explanations	Booking requirements hidden in collapsible text	Cancellation rules require separate page	Messaging lacks guidance on what hosts can answer

The table above provides an overview of all violations I identified across the tasks and heuristics. Below are detailed reports for the most significant issues, including severity ratings, specific locations, and suggested fixes. Each violation includes a screenshot to show exactly what the issue looks like.

P1. Filters refresh with no visible loading indicator

Heuristic Violated: #1: Visibility of System Status

Severity: 2 (Minor)

Location: Search results page immediately after applying filters

Description: When I tap “Apply” after selecting filters, the results update but there’s no loading spinner or status message. For a moment I’m not sure if the filter actually worked or if I need to tap the button again. The results eventually change, so it doesn’t block completion, but clearer feedback would make the process feel more responsive.

Suggested Fix: Add a brief loading indicator or “Updating results...” message while the filters are applied.

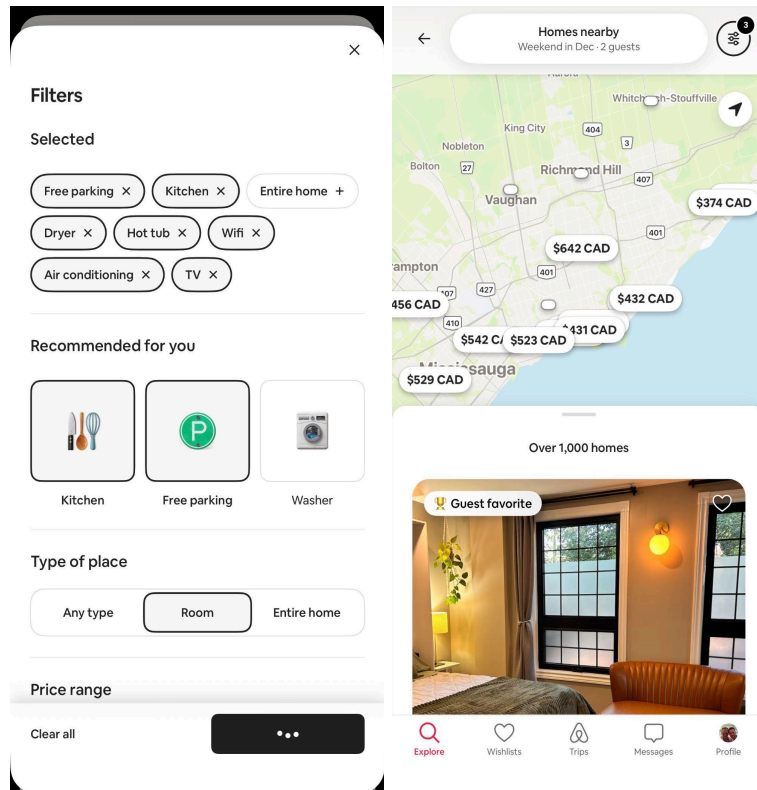


Figure 3: P1 - Filter application with no visible loading feedback

P2. Ambiguous icons in the filter panel

Heuristic Violated: #2: Match Between System and Real World

Severity: 2 (Minor)

Location: Filter menu showing icons such as “Type of place”

Description: Some of the icons used for filter categories (like “Type of place”) don’t clearly communicate their meaning without the text label. If I only saw the icon, I wouldn’t know what the option referred to. It’s not a major issue since labels are present, but the icons themselves aren’t very intuitive.

Suggested Fix: Use more recognizable icons or add tooltips/brief descriptions that appear on tap or long press.

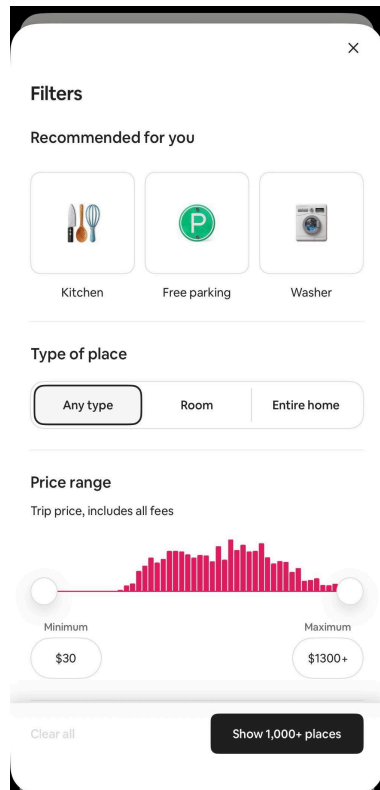


Figure 4: P2 - Ambiguous filter icons without clear meaning

P3. Losing listing scroll position when going back

Heuristic Violated: #3: User Control and Freedom

Severity: 3 (Major)

Location: Navigating back from an individual listing to the search results

Description: After viewing a listing, tapping back resets the results page to the top instead of returning to where I was. This makes it harder to compare multiple listings and I end up re-scrolling repeatedly. It becomes confusing to remember which listings I've already viewed.

Suggested Fix: Preserve the user's scroll position so the results page stays exactly where the user left it.

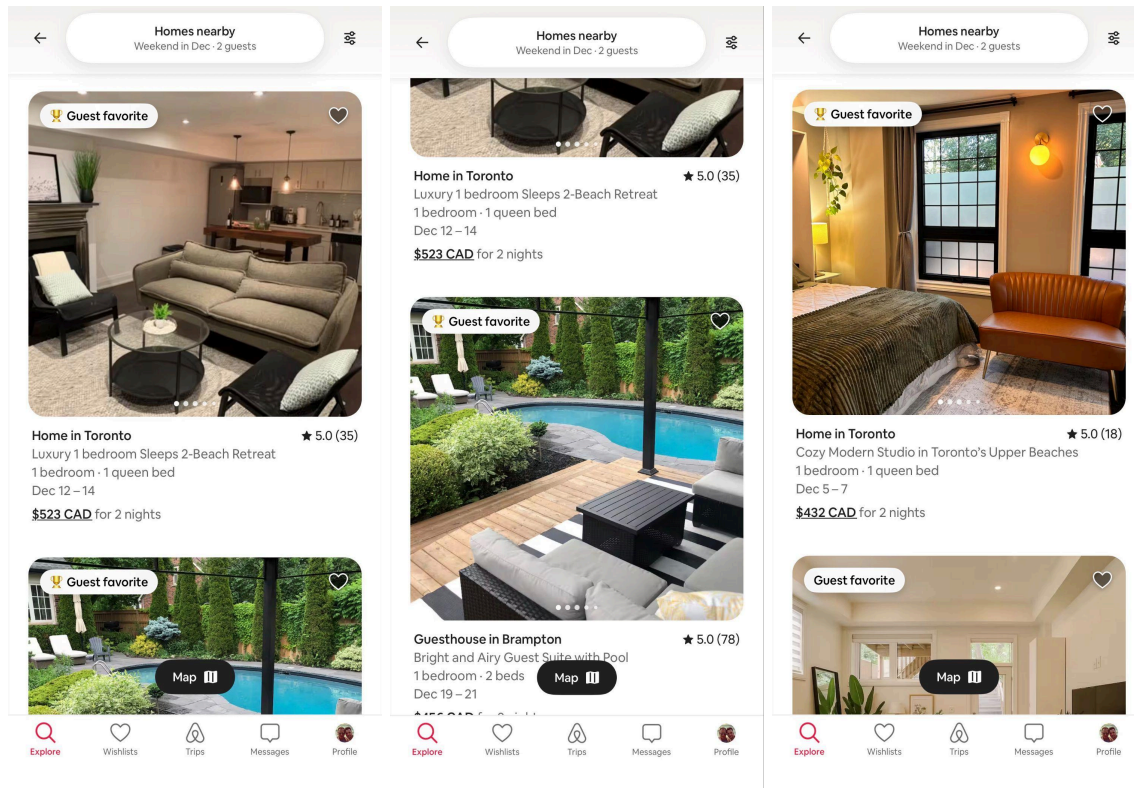


Figure 5: P3 - Scroll position lost when navigating back from listing (left: scrolled down to item #5, center: scrolled further to item #6, right: reset to top after returning)

P4. Inconsistent amenity layout across listings

Heuristic Violated: #4: Consistency and Standards

Severity: 2 (Minor)

Location: Amenities section on different listing pages

Description: Amenities are displayed differently from listing to listing. Some group them under icons while others use long text blocks. Because of this inconsistency, I had to search around each time to find the amenities section.

Suggested Fix: Use a standardized amenity layout and consistent placement across all listings.

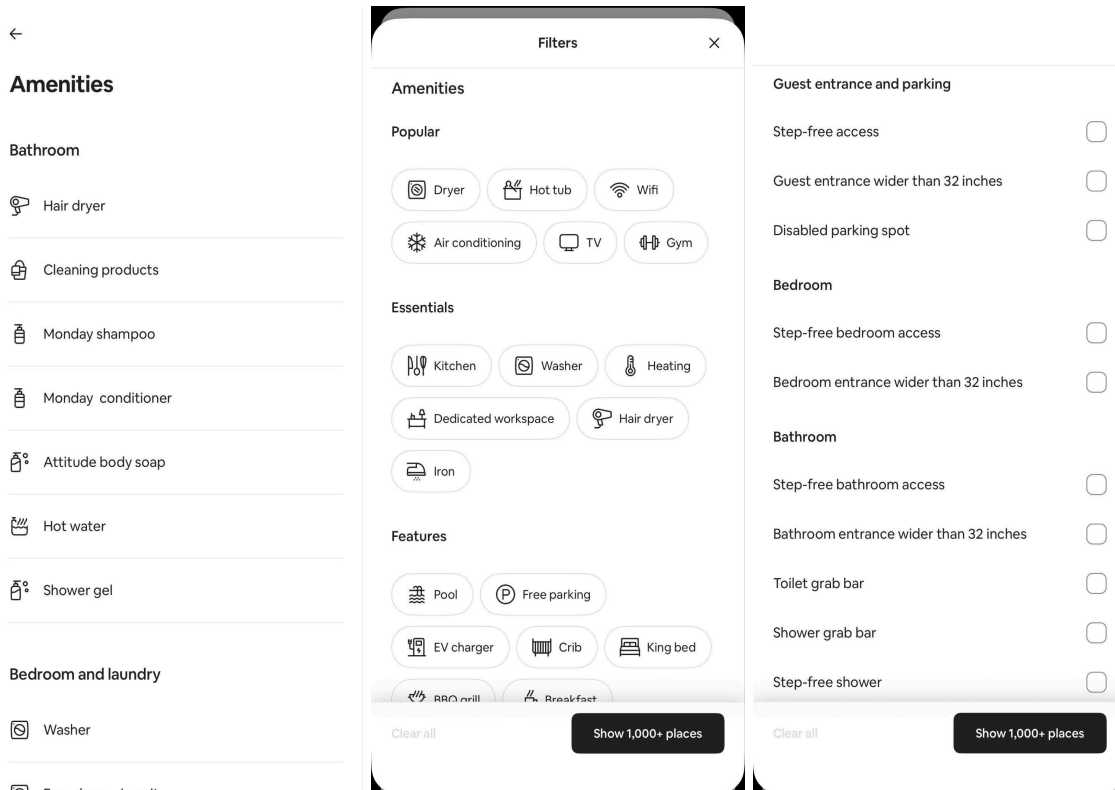


Figure 6: P4 - Inconsistent amenity layouts across different listings (left & center: icons with categorized layout from Listing A, right: text-only checkbox format from Listing B)

P5. Mis-taps on the date picker due to small touch targets

Heuristic Violated: #5: Error Prevention

Severity: 3 (Major)

Location: Calendar date picker

Description: The individual date cells are small and very close together, which caused me to tap the wrong date multiple times. It slowed me down and felt frustrating, especially when picking dates near the edge of the week.

Suggested Fix: Increase the size of tap targets or spacing between dates. A confirmation step before finalizing selected dates could also help.

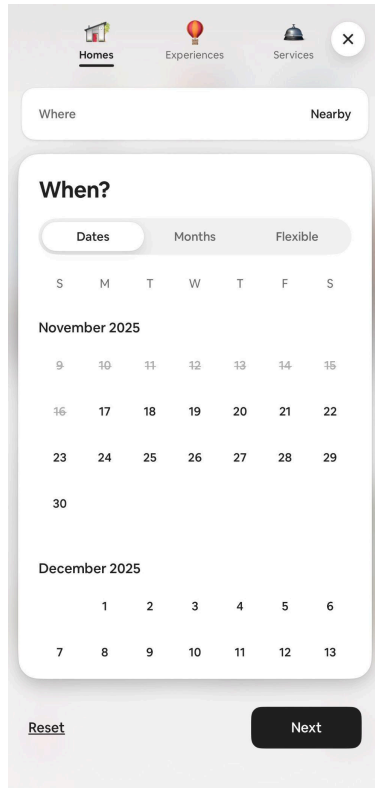


Figure 7: P5 - Date picker with small, tightly-spaced touch targets

P6. Important listing details require extensive scrolling

Heuristic Violated: #6: Recognition Rather Than Recall

Severity: 2 (Minor)

Location: Middle sections of listing pages

Description: House rules, amenities, and cancellation policies are buried far down the page. I had to remember or re-scroll to find details I saw earlier when reviewing the booking section. Important information should be easier to access.

Suggested Fix: Add a summary section or anchor links at the top for fast navigation to key details.

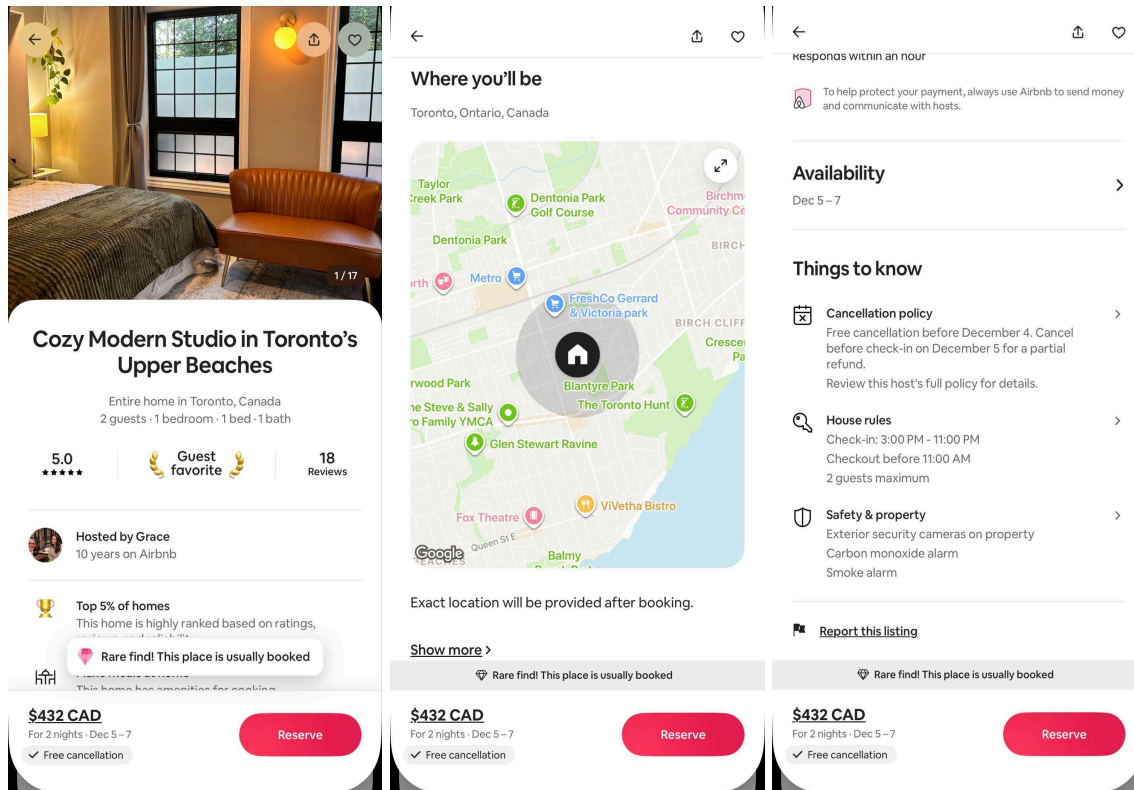


Figure 8: P6 - Important details buried deep in listing page requiring excessive scrolling (top: header and basic info, middle: location section, bottom: critical cancellation policy and house rules)

P7. No quick way to compare multiple listings

Heuristic Violated: #7: Flexibility and Efficiency of Use

Severity: 2 (Minor)

Location: Search results page

Description: The app doesn't offer any way to compare listings side-by-side. After checking several properties, it becomes hard to remember which listing had which features. This slows down the decision process.

Suggested Fix: Add a "Compare" feature that lets users select multiple listings and see key differences in a side-by-side layout.

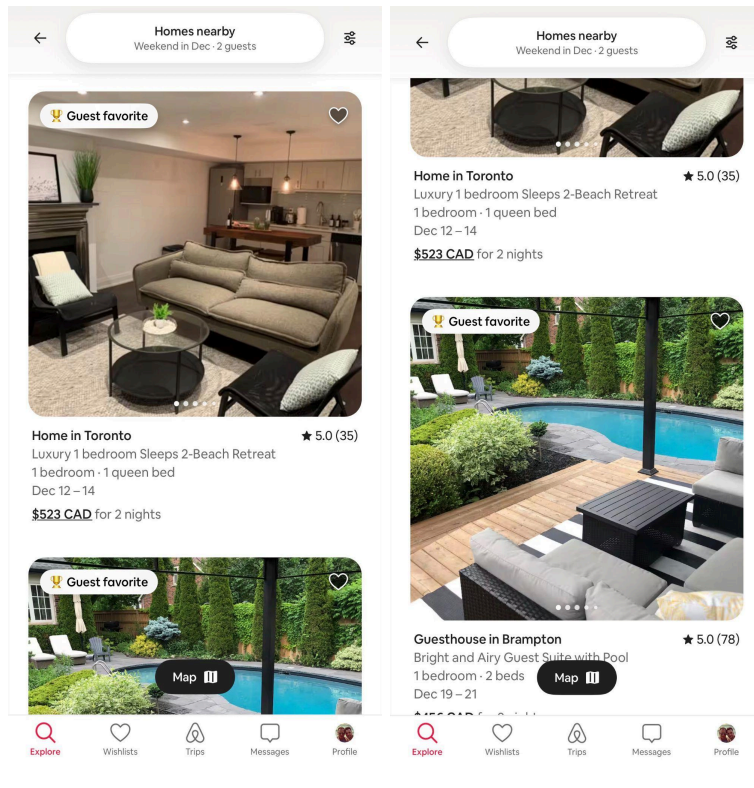


Figure 9: P7 - Search results with no listing comparison feature

P8. Listing page visually cluttered

Heuristic Violated: #8: Aesthetic and Minimalist Design

Severity: 1 (Cosmetic)

Location: Full listing page (top to mid scroll)

Description: Listing pages contain a lot of dense information in a single long scroll including photos, maps, host info, reviews, rules, and amenities. It feels visually overwhelming when trying to quickly scan for specific details.

Suggested Fix: Group related sections more clearly, increase white space, or use tabs to separate content categories.

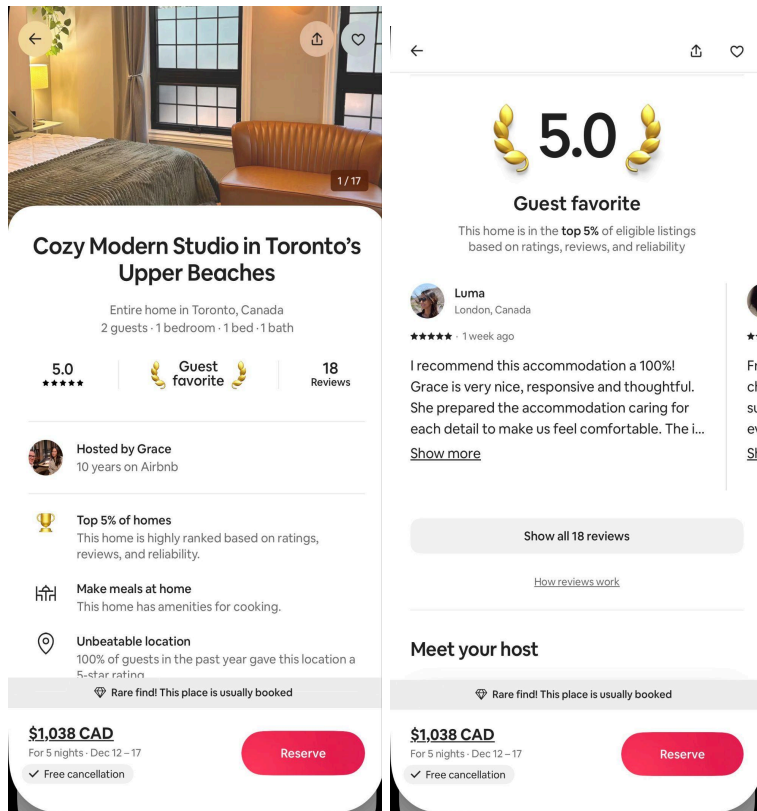


Figure 10: P8 - Visually cluttered listing page with dense information layout

P9. Vague error messages when a listing fails to load

Heuristic Violated: #9: Help Users Recognize, Diagnose, and Recover from Errors

Severity: 2 (Minor)

Location: Listing page during slow network or failed load

Description: When a listing doesn't load, the app only displays "Something went wrong," without explaining why or what to do. It's unclear whether the issue is with my connection, the app, or the listing itself.

Suggested Fix: Provide more specific, actionable error messages such as "Check your internet connection" or "This listing may no longer be available. Tap to retry."



Figure 11: P9 - Vague error message without explanation or recovery guidance

P10. No guidance on how to use advanced filters

Heuristic Violated: #10: Help and Documentation

Severity: 1 (Cosmetic)

Location: Filter categories like “Essentials” or “Accessibility features”

Description: Some filter categories use vague or specialized terms. Without explanations, it’s unclear what will be included or excluded when selecting them.

Suggested Fix: Add small info icons or short explanations that clarify what each filter option covers.

Guest entrance and parking

- Step-free access ☐
- Guest entrance wider than 32 inches ☐
- Disabled parking spot ☐

Bedroom

- Step-free bedroom access ☐
- Bedroom entrance wider than 32 inches ☐

Bathroom

- Step-free bathroom access ☐
- Bathroom entrance wider than 32 inches ☐
- Toilet grab bar ☐
- Shower grab bar ☐
- Step-free shower ☐

Clear all Show 1,000+ places

Figure 12: P10 - Advanced filter categories lacking explanatory guidance

P11. Booking flow back button exits unexpectedly

Heuristic Violated: #3: User Control and Freedom

Severity: 3 (Major)

Location: Any step in the booking flow before payment

Description: Tapping the back button sometimes exits the entire booking process instead of taking me back one step. This caused me to lose progress and restart the booking several times.

Suggested Fix: Ensure consistent back-navigation that always goes one step back. Add a confirmation prompt when exiting the flow entirely.

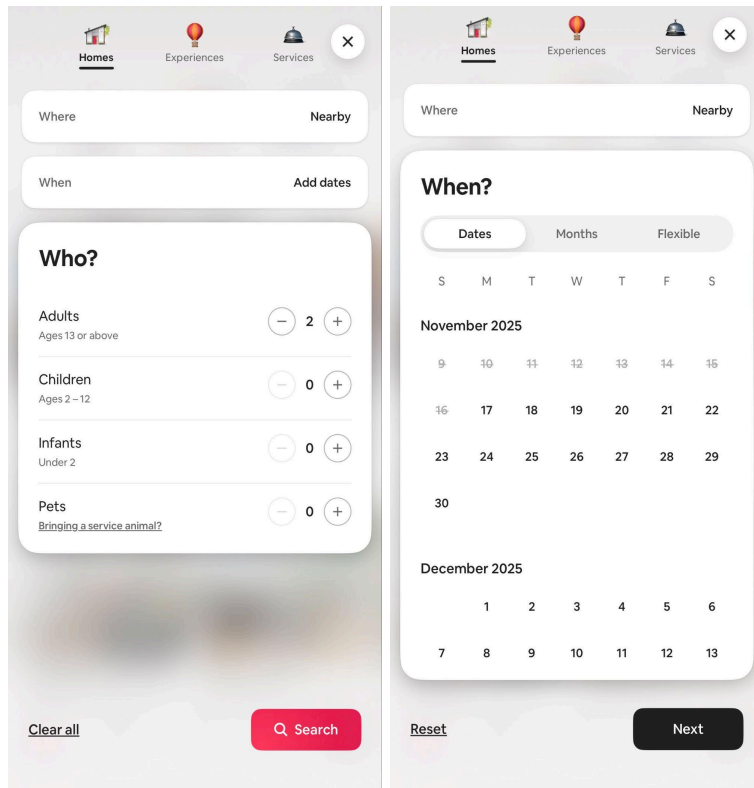


Figure 13: P11 - Back button in booking flow with inconsistent navigation behavior

P12. Price breakdown loads slowly with no feedback

Heuristic Violated: #1: Visibility of System Status

Severity: 2 (Minor)

Location: Price details section during booking

Description: When expanding the price breakdown, there's a noticeable delay but no loading indicator. It's unclear whether the tap registered or if the screen froze.

Suggested Fix: Add a small loading animation or skeleton placeholder while the breakdown loads.

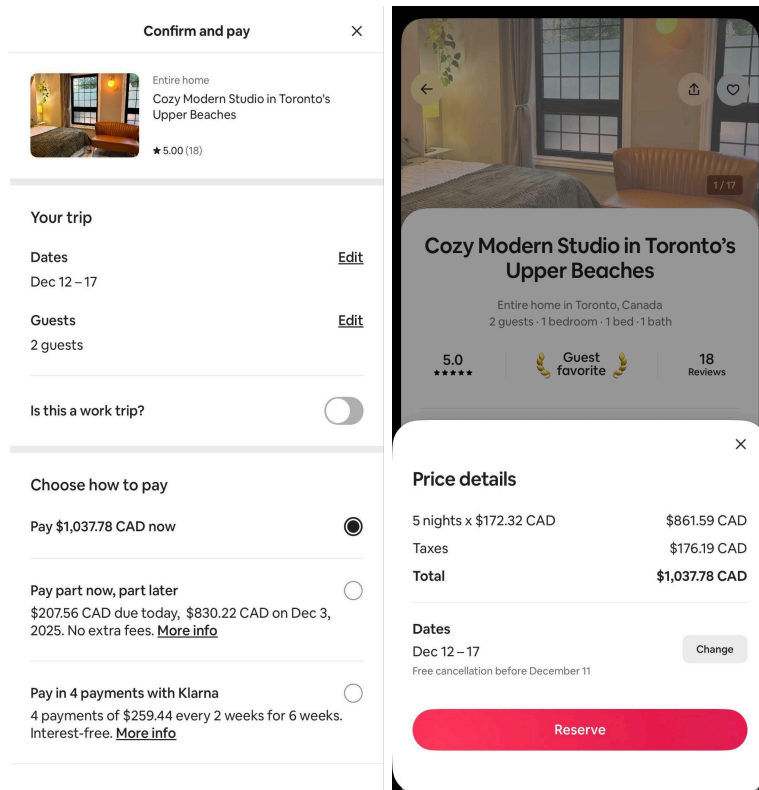


Figure 14: P12 - Price breakdown loading with no status feedback

P13. Cancellation options appear too close together

Heuristic Violated: #5: Error Prevention

Severity: 3 (Major)

Location: Reservation details → cancellation options

Description: “Cancel reservation” and “Change dates” appear close together and use similar visual styles. It’s easy to tap the wrong one, which is especially risky for irreversible actions like cancellations.

Suggested Fix: Increase spacing, use distinct button styles, or require confirmation before canceling.

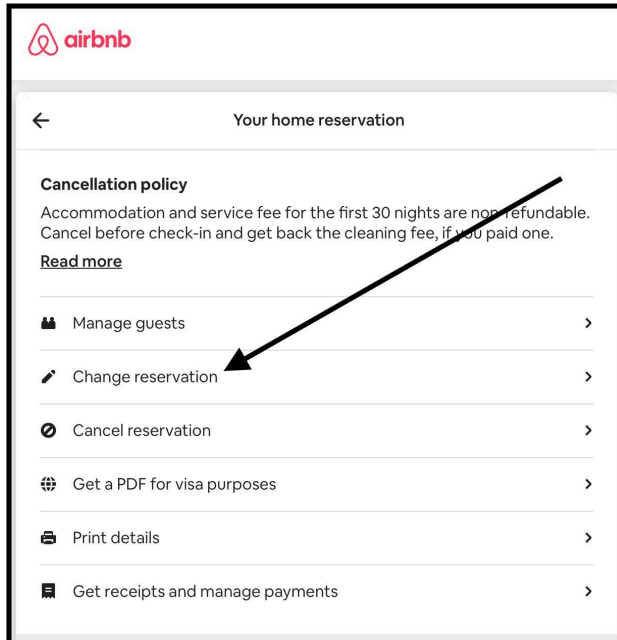


Figure 15: P13 - Cancellation options placed too close together with similar styling

P14. Message send failures lack explanation

Heuristic Violated: #9: Help Users Recognize, Diagnose, and Recover from Errors

Severity: 3 (Major)

Location: Messaging interface when offline or connection drops

Description: When I tried sending a message to a host and it failed (probably because my connection dropped), the app only showed a red exclamation mark with no explanation. I had to guess the cause and figure out on my own to try resending it. This is especially problematic because messaging is critical for coordinating check-in details and asking important questions about the property.

Suggested Fix: Add a tooltip or alert explaining the failure, such as “Message not sent — check your connection and tap to retry.”

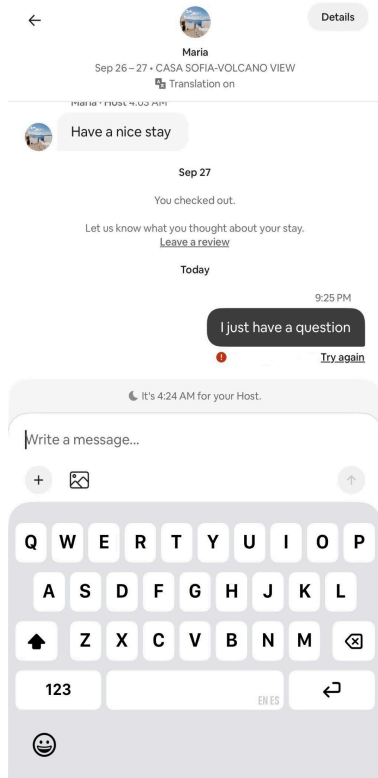


Figure 16: P14 - Message send failure with no explanation or recovery guidance

5.3) Issues-Evaluators Matrix

Since I was the only evaluator, the matrix below includes just one evaluator column. Each issue listed was identified during my independent walkthrough.

Issues-Evaluators Matrix

Issue ID	Unique Issue Description	Evaluator 1
P1	Filters refresh with no visible loading indicator	✓
P2	Ambiguous filter category icons	✓
P3	Scroll position lost after returning from listing	✓
P4	Inconsistent amenity layouts across listings	✓

P5	Small date picker touch targets causing mis-taps	✓
P6	Important details require excessive scrolling	✓
P7	No listing comparison feature	✓
P8	Listing page visually cluttered	✓
P9	Vague listing load error messages	✓
P10	No guidance for advanced filters	✓
P11	Back button exits booking flow	✓
P12	Price breakdown loads with no indicator	✓
P13	Cancellation options too close together	✓
P14	Message send failures lack explanation	✓

6. Condensed Issue List

P1 – No loading indicator after applying filters (Severity 2)

P2 – Ambiguous filter icons (Severity 2)

P3 – Scroll position resets when returning from a listing (Severity 3)

P4 – Inconsistent amenity layouts (Severity 2)

P5 – Date picker has small touch targets (Severity 3)

P6 – Key listing details require excessive scrolling (Severity 2)

- P7 – No way to compare listings (Severity 2)
- P8 – Listing page visually cluttered (Severity 1)
- P9 – Vague load error messages (Severity 2)
- P10 – Lack of help for advanced filters (Severity 1)
- P11 – Back button unexpectedly exits booking flow (Severity 3)
- P12 – Price breakdown loads slowly with no feedback (Severity 2)
- P13 – Cancellation and change-date options placed too close (Severity 3)
- P14 – Message send failures lack explanation (Severity 3)

7. Prioritization and Recommendations

7.1) Top Issues Summary (Severity 3–4)

Top Issues

Issue ID	Description	Severity	Estimated Fix Effort (1–5)	Priority Score
P13	Cancellation options too close together (risk of accidental cancellation)	3	1	3.0
P3	Losing scroll position when navigating back from a listing	3	2	1.5
P14	Message send failures lack explanation; no recovery path	3	3	1.0
P11	Back button	3	4	0.75

	exits full booking flow unexpectedly			
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7.2) Priority Ranking and Actionable Recommendations

Based on the calculated priority scores, the following order represents the recommended sequence for addressing the highest-severity issues.

Recommendation 1: Improve spacing and styling for destructive actions

Linked issue: P13, cancellation options too close together

Priority score: 3.0, highest priority

Suggested change: Increase spacing between “Cancel reservation” and “Change dates,” style destructive actions with a clearer visual identity, and add a confirmation step before executing a cancellation.

Expected impact: Reduces the risk of accidental cancellation, prevents serious user mistakes, and improves confidence during reservation management.

Recommendation 2: Preserve scroll position between listing views

Linked issue: P3, scroll position lost between listing views

Priority score: 1.5

Suggested change: Save the user’s exact scroll position when they open a listing, restore that position instantly when they return to results, and optionally mark previously viewed listings with a subtle “Viewed” indicator.

Expected impact: Improves comparison shopping, reduces repeated scrolling, and helps users keep track of listings they have already reviewed.

Recommendation 3: Provide clear error messages for messaging failures

Linked issue: P14, message send failures lack explanation

Priority score: 1.0

Suggested change: Add a tooltip or alert explaining why a message failed, suggest recovery steps such as checking the connection, and allow users to manually retry sending the message.

Expected impact: Improves trust, reduces communication uncertainty, and supports users during important guest-host interactions such as check-in coordination or property questions.

Recommendation 4: Make booking-flow back navigation consistent

Linked issue: P11, back button exits full booking flow unexpectedly

Priority score: 0.75

Suggested change: Ensure the back button moves users to the previous step instead of exiting the full booking flow. If the user attempts to leave the booking process entirely, display a confirmation dialog and optionally save progress so they can resume.

Expected impact: Prevents loss of progress, reduces frustration, and creates a more predictable booking experience.

Optional Low-Effort Enhancements

Recommendation 5: Add lightweight loading indicators for price breakdowns

Linked issue: P12, price breakdown loads slowly with no feedback

Suggested change: Use a skeleton loader, spinner, or short loading message while fee details are being fetched.

Expected impact: Improves visibility of system status and reduces confusion during checkout.

Recommendation 6: Increase whitespace in listing layouts

Linked issue: P8, listing page visually cluttered

Suggested change: Add more spacing between major listing sections and group related information more clearly.

Expected impact: Improves scanability, reduces visual overload, and helps users find key listing information more efficiently.

8. Conclusion

This usability inspection identified a range of opportunities to improve clarity, efficiency, and error prevention within the Airbnb mobile booking experience. Addressing the highest-priority issues would minimize user frustration, support decision-making, and strengthen trust during key interactions such as booking and host communication. Together, these improvements would meaningfully enhance the overall experience for Airbnb's large base of mobile users.